

Jieun Kim

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RESEARCH INTEREST

My research focuses on **vision**, **multimodal learning**, and **trustworthy AI (XAI)**. I'm particularly interested in understanding how models perceive and reason about the visual world, and how we can ensure their outputs are interpretable and reliable.

EDUCATION

- | | |
|---|---|
| • Yonsei University
Ph.D., Artificial Intelligence (Expected Graduation: Feb. 2028) | Mar.2024 - Present
Seoul, Korea |
| • Keimyung University
M.S., Computer Engineering | Mar.2021 - Feb.2023
Daegu, Korea |
| • Keimyung University
B.S., Computer Engineering | Mar.2017 - Feb.2021
Daegu, Korea |

SELECTED PUBLICATIONS

C=CONFERENCE, J=JOURNAL

- [C.9] **Spatial Reasoning in Large Vision-Language Models via Egocentric-to-Allocentric Transformation**
Yoonji Kim*, **Jieun Kim*** and Sung-Bae Cho
EMNLP Findings, Oct. 2026 (First Author)
- [C.8] **Injecting Context via Situation Working Memory for Logical Reasoning with LLMs**
Jieun Kim, Seoha Lim, Young-Hae Choi and Sung-Bae Cho
ACL, July 2026 (First Author)
- [C.7] **Diagnosing Spatial Consistency across Perspectives and Viewpoints in Large Vision-Language Models**
Yoonji Kim, **Jieun Kim**, Yujin Jeong and Sung-Bae Cho
ACL, July 2026
- [C.6] **Adaptive Beam Search with Shannon Entropy for Data-centric Reasoning in LLMs**
Yoonji Kim*, Yujin Jeong*, **Jieun Kim** and Sung-Bae Cho
PAKDD, Jun. 2026
- [C.5] **Hierarchical Prototype Learning for Semantic Segmentation**
Seoha Lim*, Jinmyeong Kim*, **Jieun Kim** and Sung-Bae Cho
ICLR, Apr. 2026
- [C.4] **Visual-Linguistic Abductive Reasoning with LLMs for Knowledge-based Visual Question Answering**
Jieun Kim, Yujin Jeong and Sung-Bae Cho
EACL, Mar. 2026 (First Author)
- [C.3] **Fuzzy Contrastive Decoding to Alleviate Object Hallucination in Large Vision-Language Models**
Jieun Kim, Jinmyeong Kim, Yoonji Kim and Sung-Bae Cho
ICCV, Oct. 2025 (First Author)
- [C.2] **Estimation of a Relative Camera Orientation with Few Correspondences Using Unsynchronized Viewpoints**
Jieun Kim and Deokwoo Lee
MITA, Jul. 2023 (First Author)
- [C.1] **Recognition of Facial Expression Using Spatial Transformation Network and Convolutional Neural Network**
Jieun Kim, Eung-Joo Lee and Deokwoo Lee
Pattern Recognition and Tracking XXXIII, May 2022 (First Author)
- [J.1] **Visual Question Answering: A Survey of Methods, Datasets, Evaluation, and Challenges**
Byeong Su Kim, **Jieun Kim**, Deokwoo Lee and Beakcheol Jang
ACM Computing Surveys, Vol. 57, No. 10, pp. 1–35, 2025 (IF 23.8)

*Equal contribution

RESEARCH & WORK EXPERIENCES

- **Research Student** Dec 2024 – Present
Soft Computing Lab, Yonsei University, Korea
 - **Focus: Trustworthy AI (XAI)** for multimodal and agentic systems
 - **Multimodal Model**
 - * **Hallucination mitigation** — fuzzy rule-based contrastive decoding [C.3]
 - * ongoing: attention-based steering, co-occurrence bias removal [ongoing]
 - * **Logical reasoning** — abductive reasoning, situation working memory [C.4, C.8]
 - * **Representation learning** — hierarchical prototype learning [C.5]
 - * **Spatial reasoning** — perspective/viewpoint consistency [C.7, C.9]
 - **Multi-Agent LLM Systems**
 - * **Orchestration** — BDI-based multi-agent framework [ongoing]
 - * **Failure attribution** — error localization in agent trajectories [ongoing]
- **Research Student** Mar 2023 – May 2023
AndLab, Korea
 - **Visual Question Answering** — survey, *ACM Computing Surveys* [J.1]
- **Research Student** Jan 2021 – Feb 2023
ISIP Lab, Korea
 - **Facial expression recognition** — spatial transformer networks, CNNs [C.1]
- **Research Assistant** Jan 2022 – Dec 2022
DGIST, Korea
 - **Camera calibration** — relative orientation, unsynchronized viewpoints [C.2]

LEADERSHIP

- **Leader of Neuro-Symbolic Research Group** Feb 2024 - Present
Soft Computing Lab, Yonsei University
 - **Mentoring** — junior researchers, acceptances at *ICLR 2026*, *PAKDD 2026*, *ACL 2026*
- **Women Graduate Student Engineering Research Team Program** May 2022 - Oct 2022
Korea Women in Science and Technology Foundation, Korea
 - **Principal Investigator** — Korean Visual Question Answering (VQA) model

REGISTERED SOFTWARE (COPYRIGHT)

- **Working Memory for LLM Agent** 2025
Korea Copyright Commission (No. C-2025-038246)
- **Dual LLM Improving Symbolic Logic for Knowledge-based Visual Reasoning Model** 2024
Korea Copyright Commission (No. C-2024-035398)
- **LoCoT: Logical Chain of Thought Reasoning for Visual Language Model** 2024
Korea Copyright Commission (No. C-2024-035781)

SKILLS

- **Programming:** Python, C, C++, C#, Kotlin, JavaScript
- **Frameworks:** PyTorch, TensorFlow
- **Research Methods:** LoRA fine-tuning, multi-agent LLM orchestration, attention-based interpretability probing, LVLM benchmark evaluation
- **Languages:** Korean (Native), English

ACADEMIC SERVICE

- **Reviewer:** *NeurIPS 2026*